

Towards Social Interaction between 1st and 2nd Person Perspectives on Bodily Play

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Figure 1: Two players, Alice and Rabbit, with the first- and second-perspectives are playing in bodily play which is a collaborative and asymmetric game. Each of them swings a balance chair in the physical space to rotate a virtual maze and a glass ball is rolled based on the laws of physics. Their common goal is to move the glass ball to an exit. (a, b) The Alice player with the first-person perspective is inside the glass ball and navigates in the maze; (c, d) The Rabbit player with a bird view assists the Alice player. Both players complement each other, and communicate and share their feelings.

ABSTRACT

Bodily play, which is a productive social interaction for bonding social relationships, has positive impacts on self-efficacy, acute cognitive benefit, and emotion. However, most bodily play encourages players to enjoy their own experiences. There are limited researches on sharing players' perspectives to enhance players' empathy for understanding others. Thus, we propose an asymmetric two-person game in an immersive environment. This bodily play, which supports perspective-taking via the integration with the first- and second-perspectives, has a collaborative interface that allows users to share their physiological and emotional perspectives. Initial testing of the system shows that players can not only understand well the feeling and problems encountered by each other through sharing perspectives and information but also increases the social closeness of players and stimulates empathy after the interplay.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction paradigms**; • **Collaborative interaction**;

KEYWORDS

Bodily play, social interaction, collaborative exertion game, empathy, person perspectives, virtual reality

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1 INTRODUCTION

In psychology theory, empathy helps humans understand the thoughts and feelings of others for harmonious interaction [1], and plays an important role in social relationships [2]. Previous studies report that collaboration activities have shown promise in enhancing empathy [3], as well as the more conditions for collaboration or positive interaction, the more likely it is to elicit stronger empathy [4]. Especially the collaborative bodily play, which has rich effects on human mental, physical, and social benefits [5], is

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presented to establish and maintain social closeness [6]. However, most bodily interplay encourages players to enjoy feelings of control over their bodies. There are limited researches on investigating players' empathy while players share their perspectives and understand others. Therefore, we design a two-person bodily play to probe using shared physiological and emotional perspectives for enhancing social interaction and empathy in the collaborative tasks. Figure 1 shows the application. The proposed system supports the performance of perspective-taking in a cycle through interdependent gameplay. The players perform bodily collaborative exertion. During the process, communication is necessary. Players exchange opinions and concern others' feelings, thereby achieving perspective-taking gradually.

2 RELATED WORK

Most bodily play supports the interplay that can lead to entertainment and bodily interactions of people [7]. In interdependent exertion, players influence each other's advancement [8], and can use shared objects [9] and space [10], or their bodies [11]. The 1st person perspective of the player's body focuses on how to enhance immersion and presence for the illusion of being there [12], including the features such as sensing the feelings of control over their body and feeling their virtual body [13]. Although they enjoy such experiential space that is around one's own body, there is a lack of sharing the self-experience feelings with others. Studies present that empathy can be stimulated by sharing what you feel and interacting with what you see [14]. Besides, Mueller *et al.* introduced the 2nd person social perspective in the game design, including a self/other distinction [15]. The epistemic theme is conscious that the status being duplicated belongs to another player [16]. Therefore, a player often simultaneously interplays with the body of himself or another player [17]. Although the player senses the feeling of the self, he or she has no feedback to the opposing player. Based on perspective-taking in psychology, giving feedback of the player's own perception could make other players experience the game perception that the player has gone through shortly, and provide a good interaction experience among players. We are inspired to take sharing feelings from the 1st perspective player's experience as an ingredient for enhancing the empathy of 2nd perspective collaborator.

3 SYSTEM DEVELOPMENT

We propose a collaborative gameplay using the immersive apparatus and body balance as a controller. The game background is based on the famous literature "Alice's Adventures in Wonderland". In the virtual space, Alice is sitting inside a glass ball in a garden maze, which has grassy walls on a large flat platform board. The goal is to make Alice go home from an entrance to an exit. The platform board has two rotation axes and each player can rotate one of the axes. The only way to make Alice move is to tilt the board so that the ball can roll in a certain direction.

3.1 Bodily Play Design

Figure 2 illustrates how the players swing the balance chairs to rotate the garden maze.

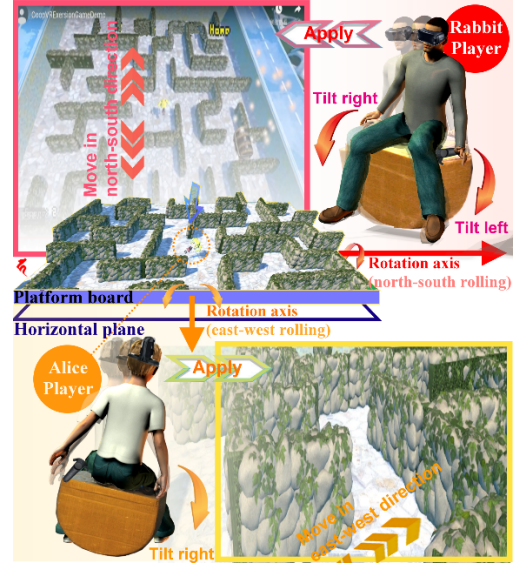


Figure 2: Alice and Rabbit players control the east-west and north-south rotation directions of the maze, respectively.

In the physical space, each player sits on a balance chair to which a VIVE controller is attached. While the player performs body swinging to the right or left side of the balance chair, the corresponding swinging amplitude is mapped to the rotation angle of the corresponding axis of the board. Both players wear a VR HMD. One player plays the role as Alice, who sees the environment on the 1st person perspective. She can see the surrounding environment and can make a good maneuver decision for locally rolling the ball. Another player, Rabbit, has a bird view on the 2nd person perspective and can estimate some reasonable paths for rolling the ball. Thus, they can drive the ball to move a desired path if they perform the exertion actions consistently.

3.2 Social Interaction Design

The goal of social interaction in game is to enhance closeness and empathy using interplay and communication. To extend the notion of perspective-taking into the bodily play domain, we adopt the asymmetrical and interdependent gameplay [8, 18-19], and take Alice as the shared object [9, 15, 17, 20]. First, the gameplay encourages physical social interplay. Alice and Rabbit players control different rolling directions of the ball, respectively. The controls of both players are complementary to each other. Therefore, the two players must work in a collaborative way so that the ball can roll along a path composed of various changes of directions. Second, both players are encouraged to share information because they complement each other for their views. The Rabbit player, who could take the role as a leader, guiding correct directions and paths for rolling the ball, also requires the surrounding information of the ball from the Alice player so that both players can make a proper action to escape from corners. Therefore, both players must communicate well for consensus decisions so that they enjoy the gameplay and successfully maneuver Alice home. Third, the Alice player who sits inside the glass ball is the shared object. She is encouraged to

share her opinions, emotions, feelings, and satisfaction feedback to the Rabbit player. Gradually, the Rabbit player with the 2nd person perspective can be inspired to care about his partner's satisfaction of gameplay. So that the Rabbit player is cultivated as an opinion recipient. Thus, we expect that the degree of closeness and empathy can be enhanced between the players.

4 CONCLUSION

We present a novel asymmetrical collaborative bodily play in an immersive virtual environment. Our work contributes to the collaborative interface that supports perspective-taking through the integration with the first- and second-perspectives, aiming at promoting social closeness and empathy for players. A preliminary version of the game was experienced by some players, and we found that players' physiological and emotional sharing information can be used as part of the interactive interface tool that makes them be more empathetic with one another. Players can understand well the feeling and problems encountered by each other through sharing perspectives and information to increase social closeness and empathy.

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